B46_EXP9.C

```c
#include<stdio.h>
#include<conio.h>
void binarysearch(int a[], int n,int val)
{
int pos =-1;
int low =0;
int high =n-1;
int mid;

while (low <= high) {
mid = (low + high)/2;

if (val == a[mid]) {
pos = mid;
printf("\n value %d found at position (index): %d\n", val,pos);
return;
}
else if(val < a[mid])
{
high = mid-1;
}
```

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```
}
else
{
low = mid+1;
}
}
}

if (pos == -1) {
printf("Value is not present in the array\n");
}
}

int main() {
int a[100];
int n,val,i;

clrscr();

printf("binary search---\n\n");
printf("enter the number of elements in array\n");
scanf("%d",&n);
```

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```
if(n<=0|| n>100) {
printf("error: invalid size");
getch();
return;
}

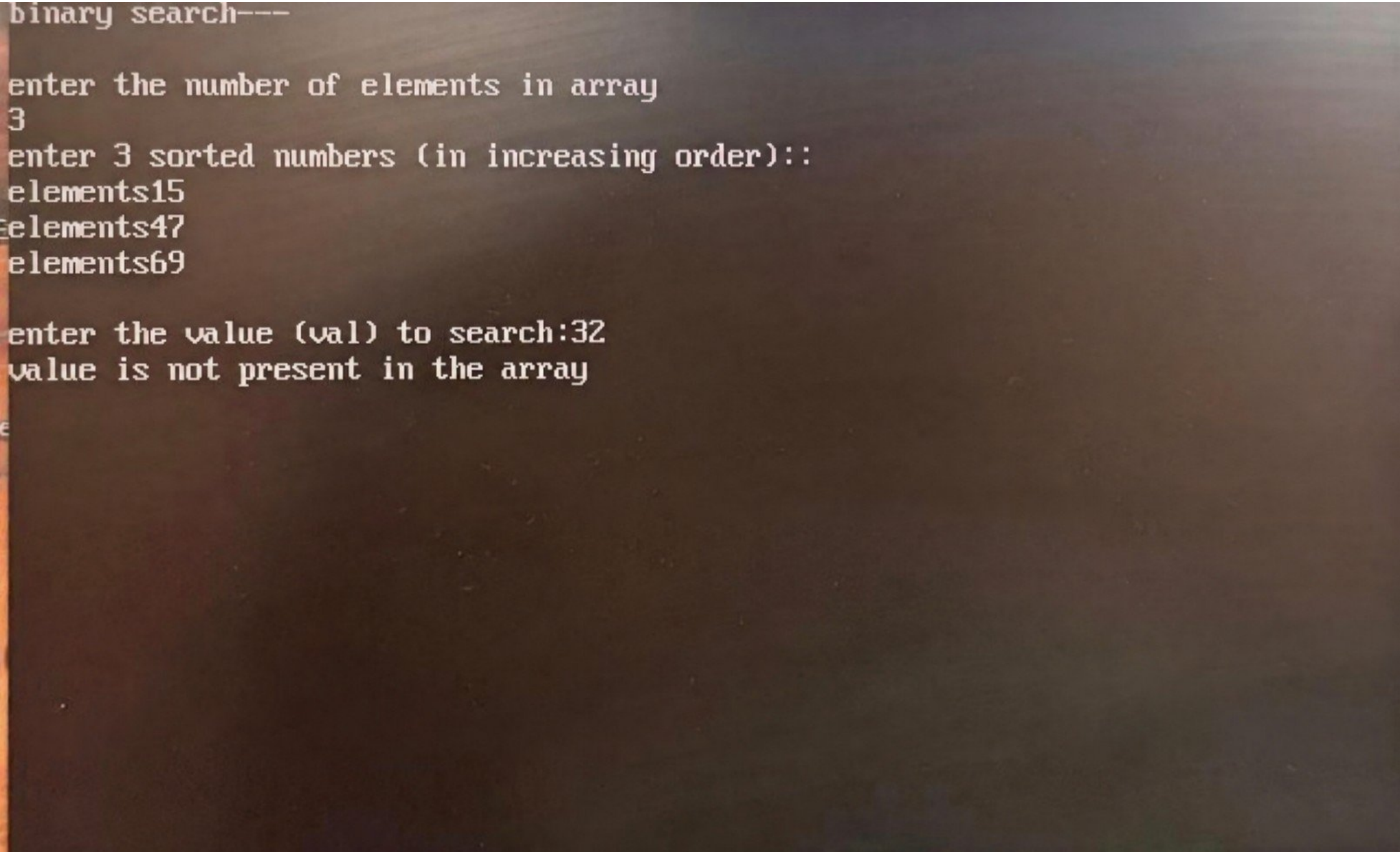
printf("enter %d sorted numbers (in increasing order)::\n",n);
for (i=0; i<n;i++) {
printf("elements",&i);
scanf("%d",&a[i]);
}

printf("\nenter the value (val) to search:");
scanf("%d",&val);

binarysearch(a,n,val);
getch();

return ;
}
```

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binary search---
enter the number of elements in array
3
enter 3 sorted numbers (in increasing order)::
elements15
elements47
elements69
enter the value (val) to search:32
value is not present in the array